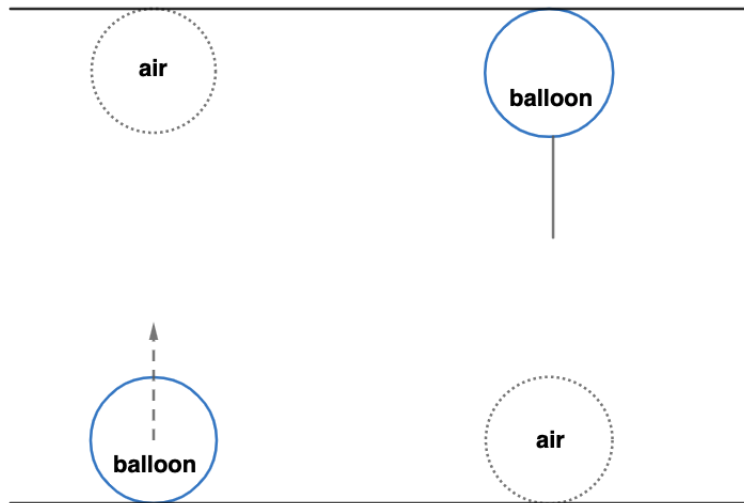


2023 F=ma Exam: Problem 9

Kevin S. Huang



Since the balloon moves from the floor to the ceiling, the air initially near the ceiling is effectively moved to the volume left by the balloon near the ground. Thus, the gravitational potential energy of the air has decreased so the answer is E.